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The campus cloud platform setup based on virtualization technology

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Abstract

In the new era, with the increase of users' demand for Internet content, Software and resources have been released and accessed as services. The service mode "IT resources should be treated as water and electricity" has put forward higher requirements for the data center construction, which determines that the traditional data center construction model and programs are no longer able to meet the needs of innovative applications of the new era. Therefore, the integration and innovation of new and old technologies is required. The high reliability, scalability, security, flexibility, greenness and resource customization of cloud computing technology enable more and more campuses to use cloud computing platforms to build campus networks. This paper mainly elaborates the use of virtualization technology and cloud computing technology to build campus cloud platform.

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1. Introduction

Over the past two decades, the Internet has connected units and individuals all over the world and realized the sharing of resources. This has also profoundly affected the various business forms and the daily life of everyone. In this new era, users' contributions to and interactions with Internet are unprecedented. Software and resources are more distributed and accessed through the Internet in the form of services. The service mode "IT resources should be treated as water and electricity" has put forward higher requirements for the data center construction, which determines that the traditional data center construction model and programs are no longer able to meet the needs of

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innovative applications of the new era. Therefore, the integration and innovation of new and old technologies is required, which is "cloud computing" technology.

At present, both vocational colleges and universities have built their own campus network information platform to carry out teaching, research, management, foreign exchange and other related work. Therefore, it will have a direct impact on the development of various vocational colleges and personnel training. Cloud computing technology is a hot new technology in today's IT industry, but also represents the trend of the future, which has largely overturned the traditional IT application mode. Its high reliability, scalability, security, flexibility, greenness, resource customization and other advantages, so it will surely bring new opportunities for the development of IT industry. The integration of cloud computing technology and data center system to build a new generation of campus cloud platform based on virtualization technology will be the new direction for the development of data center system of campus network in the future.

2. Overview of Cloud Computing

Cloud computing technology can be interpreted from different perspectives by different enterprises and individuals. The first sentence of the definition of cloud computing from Wikipedia is that Cloud computing relies on sharing of resources to achieve coherence and economies of scale, similar to a public utility. The recommended standard given by the National Institute of Standards and Technology (NIST) under the US Department of Commerce in January 2011 suggested that cloud computing should have the following characteristics:

- 1) On-demand self-service means that the customers can obtain the service on demand without the participation of service staff.
- 2) With enough bandwidth, various computing power can be provided to customers with sufficient bandwidth.
- 3) The computing power of service providers will Resource pooling will exist in the form of resource pool, providing various physical resources and virtual resources dynamically to clients in the form of multi-tenant leasing. These resources can be easily allocated or redistributed, while the customers should have a clear understanding of the location of resources.
- 4) Computing power can be distributed to customers with extremely fast speed and great flexibility. The so-called flexibility here means that customers can purchase an unlimited number of services at any time.
- 5) The service should be measurable. The cloud platform can automatically control and optimize resources.

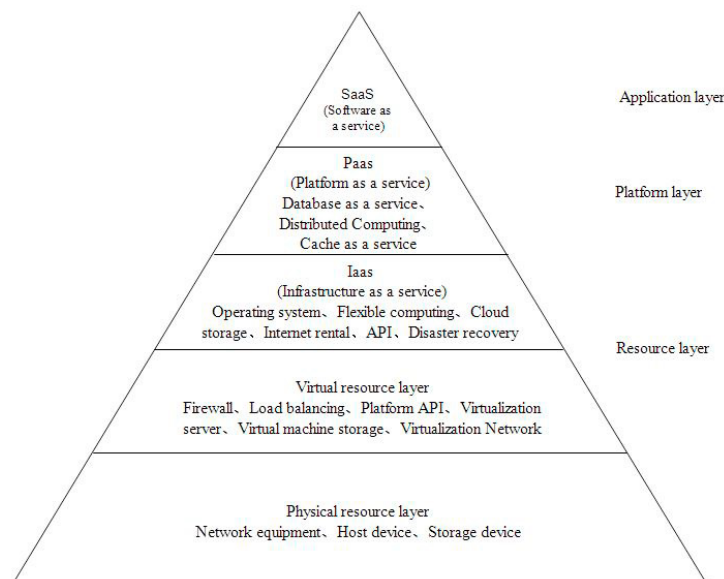


Fig. 1. Cloud computing service architecture

The cloud platform can be divided into public and private cloud based on different purposes of the service object and the purpose of setting up the cloud. According to the service content, the cloud platform is usually divided into three levels: IaaS, PaaS and SaaS. At these three levels, IaaS is at the bottom, and end-users, SaaS providers, and PaaS providers can get the computing power they need for their applications from infrastructure services, but they cannot pay the corresponding original investment costs for the fundamental IT software and hardware which support this kind of computing power.

3. Virtualization Technology

In fact, virtualization technology is a special kind of resource management technology. It breaks down the barriers of the close relationship between the traditional physical entities and abstracts and presents the traditional IT resource entities such as network, storage, computing and so on, which enables users to adopt related resources in a better way. Compared with the traditional resource management technology, virtualization technology can greatly expand the hardware capacity and simplify the software reconfiguration process, which is the best choice for cloud computing technology.

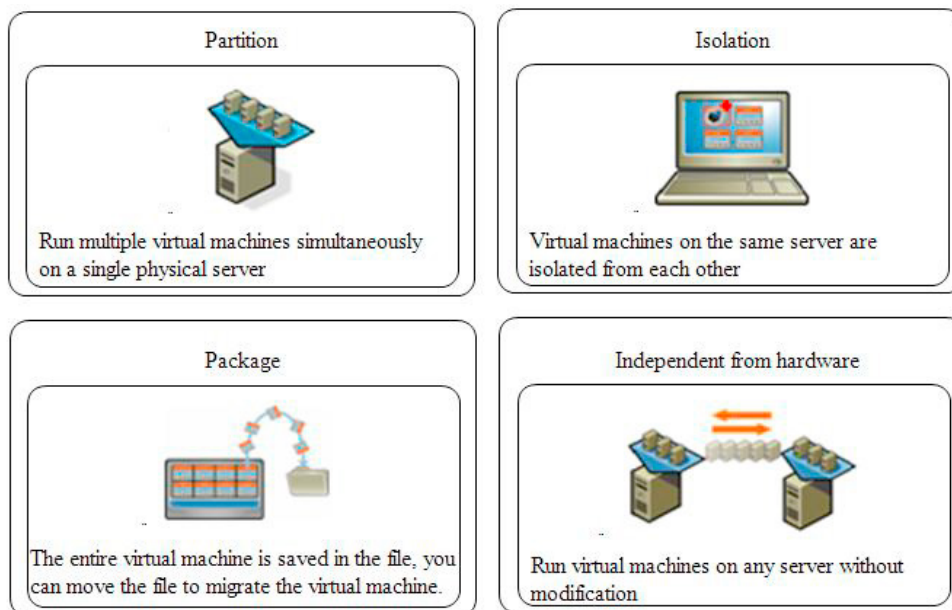


Fig. 2. The essence of virtualization technology

According to the realization mechanism of virtualization, the virtualization technologies can be divided into full virtualization, paravirtualization and hardware virtualization. The calling code of the virtual guest operating system to the underlying hardware is converted by the hypervisor in the middle. The virtual machine monitor is used to coordinate between the guest operating system and bare hardware. The full virtualization without changing the guest operating system has good compatibility, such as VMware Workstation, VSphere.

The paravirtualization technology is to add specific virtualization instructions in the virtual guest operating system and then these instructions can be directly used to call hardware resources through the Hypervisor layer. Paravirtualization requires some modification of the guest operating system so that the guest operating system realizes that it is in a virtualized environment. The typical representative is the Xen virtual machine.

Hardware-assisted virtualization technology is provided by hardware manufacturers, mainly used with the full virtualization and paravirtualization. It is an operating mode where a new instruction set and processor are added in

the CPU. Then the virtual operating system could directly call the hardware resources. The typical technology is Intel VT, AMD-V.

4. Campus network data center analysis

This paper selects a comprehensive higher vocational college with 16,000 students. The College Network Center is mainly responsible for the construction of college campus network information platform. It relies on the data center system to provide college and external users with common basic network services such as WWW, DHCP, FTP, Email, DHCP, VPN, Internet authentication, etc. Professional network services: UF financial software, educational administration system, e-books, library management system, paperless office system, student records management system, open network teaching platform, boutique resources sharing courses platform, VOD video on demand and other network services. In order to ensure all types of basic network services and professional application services provided by the campus network can operate normally, the current campus data center system adopts the traditional data center system architecture mode and does not utilize the cloud computing technology to realize the data center system virtualization. College has 10 professional servers and the relevant network applications are directly deployed on the physical server. The "one to one" or "one to many" corresponding mode is adopted between physical servers and network applications. The main network applications or services that need to consume large amounts of resources are deployed on a physical server; or multiple secondary network application services or multiple network application services that consume less resources can be centrally deployed on a physical server. To solve the problem of poor reliability caused by a single point of failure, the college deployed disk arrays on selected key applications such as the student records management system, paperless office mail system louts, to achieve local shared storage. However, with the passage of time, it is unnecessary to use disk-based disk array to achieve local shared storage and improve the reliability of the system.

With the advancement of information technology, more and more new businesses and applications will continue to be introduced into the network system. In higher vocational colleges, departments and secondary institutes will inevitably establish more systems to meet the needs of the information age. If the traditional data center system architecture model keeps being adopted, when introducing a new business, you need to configure a new web server, which brought a lot of pressure on costs and caused great resources waste. The increase of equipment will inevitably result in the decentralization of equipment, which also brings a great burden on the management of the data center system room, and the corresponding power consumption will also increase accordingly.

5. Campus cloud platform developing based on virtual technology

According to the analysis of campus data center, with the increase of business volume, traditional information system operation and maintenance mode gradually restricts the rapid development of business. Cloud computing technology based on the virtualization integrates computing, storage and network resources of data center, which can provide a common computing platform to meet the growing business needs of the campus. This paper uses zstack cloud technology as the system construction platform of data center.

5.1. Overview of Zstack

ZStack is the next generation open source cloud computing IaaS (infrastructure as a service) software. It is primarily for the smart data center of the future and manages a wide range of resources in data centers including computing, storage and networking by providing sophisticated APIs. Users can build their own intelligent data center based on ZStack or build flexible cloud application scenarios on a stable Zstack such as VDI (Virtual Desktop Infrastructure), PaaS (Platform as a Service), SaaS (Software as a Service), etc.

Zstack's single-managed node easily handles at least 30,000 concurrent API calls per second, capable of managing hundreds of thousands of physical server clusters and millions of virtual cloud hosts. ZStack's highly-effective cloud engine makes it easy to handle complex cloud computing issues. Even if the user initiates the request to create 10,000 cloud hosts, the entire creation process is only 23 minutes after ZStack effective scheduling. ZStack provides a very complete NFV (network functions virtualization). The unique distributed DHCP function avoids the

single point of failure of the network center node and ensures that the network performance of each cloud host is highly efficient and stable. It supports cloud routing function, so that you can provide each tenant a dedicated network environment. Network administrators can provide comprehensive network services (such as DHCP, DNS, SNAT, flexible IP, port forwarding, load balancing, flexible security groups, IPSec tunneling, etc.) without having to purchase additional hardware. ZStack supports the docking of various storage devices, including DAS, NAS, SAN, distributed storage, shared file system and so on, suitable for storage upgrading of traditional data centers

5.2. Hyper convergence program

Hyper-converged infrastructure (HCI) is an information infrastructure framework that integrates storage and virtual computing. With this architecture, each converged unit will become part of the storage system and virtualization system. Through a high-speed Ethernet network, a number of fusion units will be interconnected to build a hyper-converged infrastructure platform.

The campus cloud platform uses zstack super-fusion architecture and zstack control node as a cloud host, which starts in the distributed storage and provides high availability of management nodes through the storage of high reliability.

Through hyper-convergence, users can minimize the risk of business interruption in abnormal situations. As the management node runs in the virtual machine, when two nodes are available, the management node can operate normally. It ensures that the service cloud host is highly available, and can automatically switch to other converged units to run when the network or node fails, which can quickly restore the service system. Furthermore, in a data block, a hard disk or a node failure (within the failure domain definition range), it is able to ensure data availability and reliability (downgrade does not affect access).

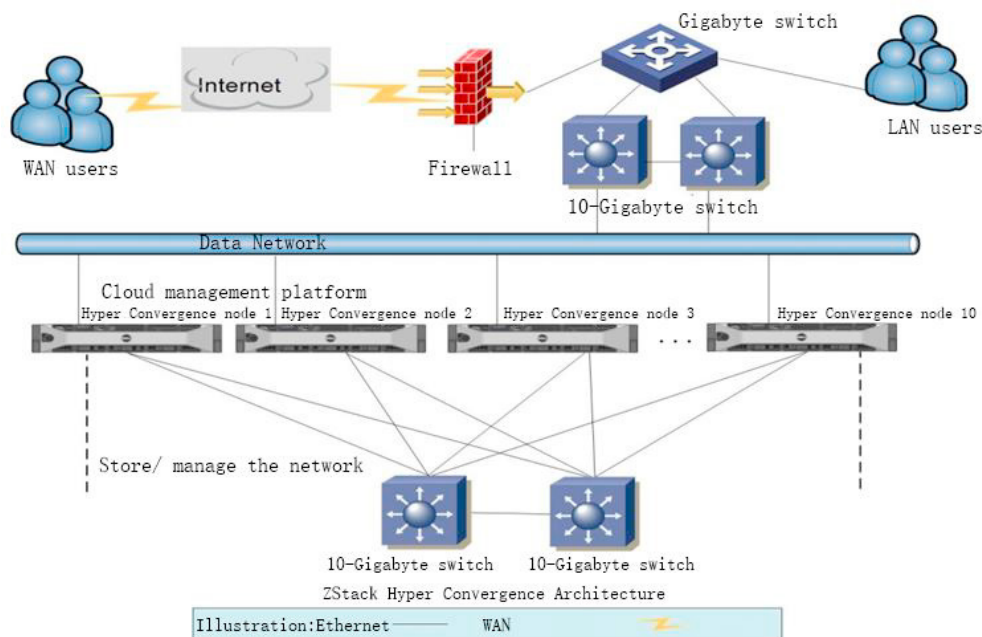


Fig. 3. Campus network hyper-integration program

Overall, hyper convergence programs ensure data security to a large extent and are flexible in capacity expansion. It is suitable for enterprise-class data center upgrades to meet enterprise-class performance computing needs and data security storage. ZStack solution has a good scale-out of the horizontal expansion capabilities. It can be extended to more than 100 large-scale clusters, thus providing a better disaster recovery guarantee.

6. Summary

The application of a cloud computing platform based on virtualization will significantly improve resource utilization of the campus data centre and system reliability while reducing maintenance difficulty and maintenance cost of the data centre. It can achieve flexible data allocation in the data centre system and enhance disaster recovery capability of the data centre.

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